



Experimental and kinetic modeling investigation on pyrolysis and combustion of *n*-butane and *i*-butane at various pressures

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ABSTRACT

Butane is the smallest alkane with normal and branched isomers. To obtain insight into the effects of fuel structure and pressure on its intermediate-to-high temperature combustion chemistry, flow reactor pyrolysis and laminar burning velocities of the butane isomers were investigated at various pressures. In the pyrolysis experiments, species profiles were measured as function of the heating temperature at 0.04, 0.2 and 1 atm using synchrotron vacuum ultraviolet photoionization mass spectrometry. Laminar burning velocities of both *n*-butane/air and *i*-butane/air mixtures were measured at 298 K and 1–10 atm using spherically expanding flames. It was observed that both the pyrolysis and combustion behaviors of the butane isomers were influenced by the fuel structures. A detailed kinetic model of butane isomers was developed and validated against the new experimental data. Both rate of production analysis and sensitivity analysis were performed to give insight into the chemistry of butane pyrolysis and combustion. In the flow reactor pyrolysis, the weaker primary-tertiary C–C bond than the primary-secondary and secondary-secondary C–C bonds leads to lower initial decomposition temperatures of *i*-butane than *n*-butane. Under both pyrolysis and combustion conditions, the reaction pathways towards C₂ and C₃ species pool are emphasized for the decomposition of *n*-butane and *i*-butane, respectively. The more abundant production of C₃ precursors explains the higher concentrations of benzene in the *i*-butane pyrolysis, while the higher laminar burning velocities of *n*-butane/air mixtures at all investigated pressures are mainly attributed to the easy production of H atom from *n*-butane decomposition and the dominance of the reactive C₂ chemistry. Moreover, the *i*-butane/air flames exhibit stronger pressure dependence than the *n*-butane/air flames. The model was further validated against a wide range of experimental data in the literature, including ignition delay times and species profiles in flow reactor pyrolysis and oxidation, shock tube pyrolysis and oxidation, jet-stirred reactor oxidation and laminar premixed flames.

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1. Introduction

Alkanes constitute an important component family in transportation fuels [1,2], playing a critical role in determining their combustion characteristics such as ignition, flashback, engine knock and combustion instabilities [3]. Among them, butane is the

smallest alkane with both normal and branched isomers, and is a major component of liquefied petroleum gas [4,5] and a minor component of natural gas [6] and syngas [7]. The butane isomers, i.e. *n*-butane and *i*-butane, consist of primary, secondary and tertiary carbon atoms and consequently three different types of C–C bonds. Thus they are widely used as prototype fuels for understanding the combustion chemistry of large alkanes [8–10].

Extensive experimental investigations have been conducted on *n*-butane and *i*-butane combustion, leading to the determination of global combustion parameters such as ignition delay times [5,8–17] and laminar burning velocities [18–30], as well as speciation in flow reactor pyrolysis [31,32] and oxidation [33], high-pressure shock tube pyrolysis and oxidation [34–36], jet-stirred reactor (JSR) oxidation [37–39] and rich premixed flames [40–

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[42]. Specifically, Dagaut et al. [37,39] investigated the JSR oxidation studies of the butane isomers at 1 atm, 1000–1450 K and equivalence ratios of 0.5, 1.0 and 1.5 using gas chromatography (GC). Oehlschlaeger et al. [43] measured the time history of the methyl radical concentration in the shock tube pyrolysis of *n*-butane/Ar and *i*-butane/Ar using laser absorption spectroscopy at 0.2–8.8 atm and 1297–1601 K, and determined the rate constants of the C–C bond dissociation reactions. Oßwald et al. [41] studied rich ($\phi = 1.71$) laminar premixed flames of *n*-butane and *i*-butane at 40 mbar using electron-impact ionization mass spectrometry (EIMS) and synchrotron vacuum ultraviolet photoionization mass spectrometry (SVUV-PIMS). Zhang et al. [32] identified the pyrolysis products of the butane isomers at a very low pressure (3 Torr) and 823–1823 K using SVUV-PIMS at the National Synchrotron Radiation Laboratory (NSRL), but without providing any concentration information. Ogura et al. [5] measured the ignition delay times of different ratios for *n*-butane/*i*-butane mixtures diluted in Ar behind reflected shock waves at 1.9–2.4 atm, 1200–1600 K, and equivalence ratio of 0.72. Healy et al. [9,10] measured the ignition delay times of *n*-butane and *i*-butane at various equivalence ratios (0.3, 0.5, 1, and 2) in a shock tube and a rapid compression machine, extending the range of pressures up to 30 atm and temperatures down to 690 K. Laminar burning velocities of *n*-butane/air and *i*-butane/air mixtures were measured at 298 K and 1 atm by several groups [18,24,26], while Farrell et al. [44] measured the values for the butane isomers at 450 K and 304 kPa. Marshall et al. [45] calculated the laminar burning velocities of *n*-butane at 1–15 atm from the pressure record in a constant volume combustion vessel. While these studies provide valuable validation data for the development of kinetic models of butane isomers [9,10,42,46], further investigations on the pyrolysis and combustion of butane isomers are still desired. Specifically, it is noted that the pyrolysis of butane isomers was performed with limited speciation information, while large discrepancies in the laminar burning velocities of butane isomers exist between the high-pressure results of Farrell et al. [44] and Marshall et al. [45], which leads to the need of more high-pressure measurements [45].

In this work, the flow reactor pyrolysis of *n*-butane and *i*-butane were studied at 0.04, 0.2 and 1 atm over the temperature range of 840–1350 K by using SVUV-PIMS to help understand the different pyrolysis behaviors between the butane isomers. The laminar burning velocities were measured from spherically expanding flames at 298 K and 1–10 atm to provide new data for exploring the high-pressure combustion characteristics of the butane isomers. A kinetic model for the butane isomers was developed and validated comprehensively against the experimental data obtained in this work, as well as the literature data from intermediate to high temperatures including species profiles and global combustion parameters. Kinetic analysis including the rate of production (ROP) analysis and sensitivity analysis has also been conducted to provide insight into the combustion chemistry influenced by the fuel structure, pressure and equivalence ratio.

2. Experimental methods

2.1. Flow reactor pyrolysis

The flow reactor pyrolysis experiment was performed at NSRL, University of Science and Technology of China (USTC). Detailed descriptions of the synchrotron VUV beamline (BL03U) are given in Ref. [47]. The schematic diagram of the pyrolysis reactor is shown in Fig. 1(a). Compared with our previous pyrolysis reactor [48–51], heating length in this reactor was extended from 150 mm to 224 mm and the density of the heating wire was doubled in region A (the first quarter heating length) and region C (the last eighth heating length). Consequently a broader high-temperature

region can be obtained at the temperature profile along the reactor centerline, as shown in Fig. 1(b). During the pyrolysis experiment, the gas flow rates of fuel (*n*-butane or *i*-butane, 99.9%) and diluent gas (argon, 99.999%) were controlled at 20 and 980 standard cubic centimeters per minute (SCCM) by mass flow controllers. According to Shrestha et al. [31], the heat of reaction in such diluted pyrolysis systems is weak and was found to have only very small influence on the centerline temperature profiles (≤ 5 K). This very small variation was also considered in the uncertainties of the measured temperatures. The fuel/Ar mixtures were then fed into a flow tube which is made of α -alumina to reduce wall catalytic effects [52,53]. Previous study [54] has shown that α -alumina is almost 1000 times less active than reforming grade γ -alumina under the same surface area, while its surface area of unit mass is an order of magnitude smaller than reforming grade γ -alumina. A small inner diameter (i.d.) of 7.0 mm of the flow tube was selected to ensure strong radial diffusion effects, reduce radial concentration gradients and achieve adequately homogeneous reaction circumstances [55–57]. The pyrolysis species were sampled at 5 mm downstream from the outlet by a quartz nozzle. The formed molecular beam was ionized by the synchrotron VUV light after entering the photoionization chamber through a nickel skimmer, followed by detection by a home-made reflected time-of-flight mass spectrometer (RTOF-MS). A series of pressures (0.04, 0.2 and 1 atm) were investigated in this work. Using the same strategy as that of Skjøth-Rasmussen et al. [59], the centerline temperature profiles of the flow tube were measured under non-reacting conditions (i.e., 1000 SCCM flow rate of argon) after experiment by an S-type thermocouple, which is not shown in Fig. 1(a). The measured centerline temperature profiles were identified as the maximum values ($T_{\max}$) [51,58] and the relationships between the mole fractions of species measured at the outlet of the flow tube and $T_{\max}$ are provided as experimental data. The uncertainty was estimated to be within ± 30 K. The data evaluation method can be found in previous work [50], and the photoionization cross sections (PICs) are available in the online database [60]. Mole fraction uncertainties were estimated to be 10%, 25% and a factor of 2 for the major species, intermediates with known PICs and those with estimated PICs, respectively.

2.2. Laminar burning velocity

Detailed descriptions of the laminar burning velocity apparatus at Shanghai Jiao Tong University (abbr. SJTU apparatus) are given in [61]. Briefly, the constant-volume, single-chamber apparatus consists of a 2.77 L cylindrical combustion vessel (150 mm i.d., 152 mm length), a 9.06 L premixing vessel, a sample inlet system, an ignition coil and a schlieren system. Mixtures of synthetic air (21% O₂ and 79% N₂) and fuel (*n*-butane or *i*-butane) were prepared using the partial pressure method in the premixing vessel. After mixing by a magnetic stirrer, the combustible gas was fed into the combustion vessel and ignited by two tungsten electrodes. Thus, a spherically expanding flame was formed and the schlieren system with a high speed camera was used to record the flame propagating process. The camera was running at 10,000 frames per seconds (fps), with a resolution of 480 × 480 pixels, such that the spatial resolution of the camera corresponded to around 0.15 mm/pixel.

Descriptions of the constant-pressure, dual-chamber cylindrical combustion vessel at Princeton University (abbr. PU apparatus) are given in Ref. [62]. Briefly, the dual-chamber vessel consists of an inner chamber (82.55 mm i.d., 127 mm length) and an outer chamber, with the wall of the inner chamber consisting of a series of holes that could be mechanically opened and closed so as to control the initial separation of the gases between the two chambers. The inner chamber was initially filled with the test combustible mixture, while the outer chamber with nitrogen to match

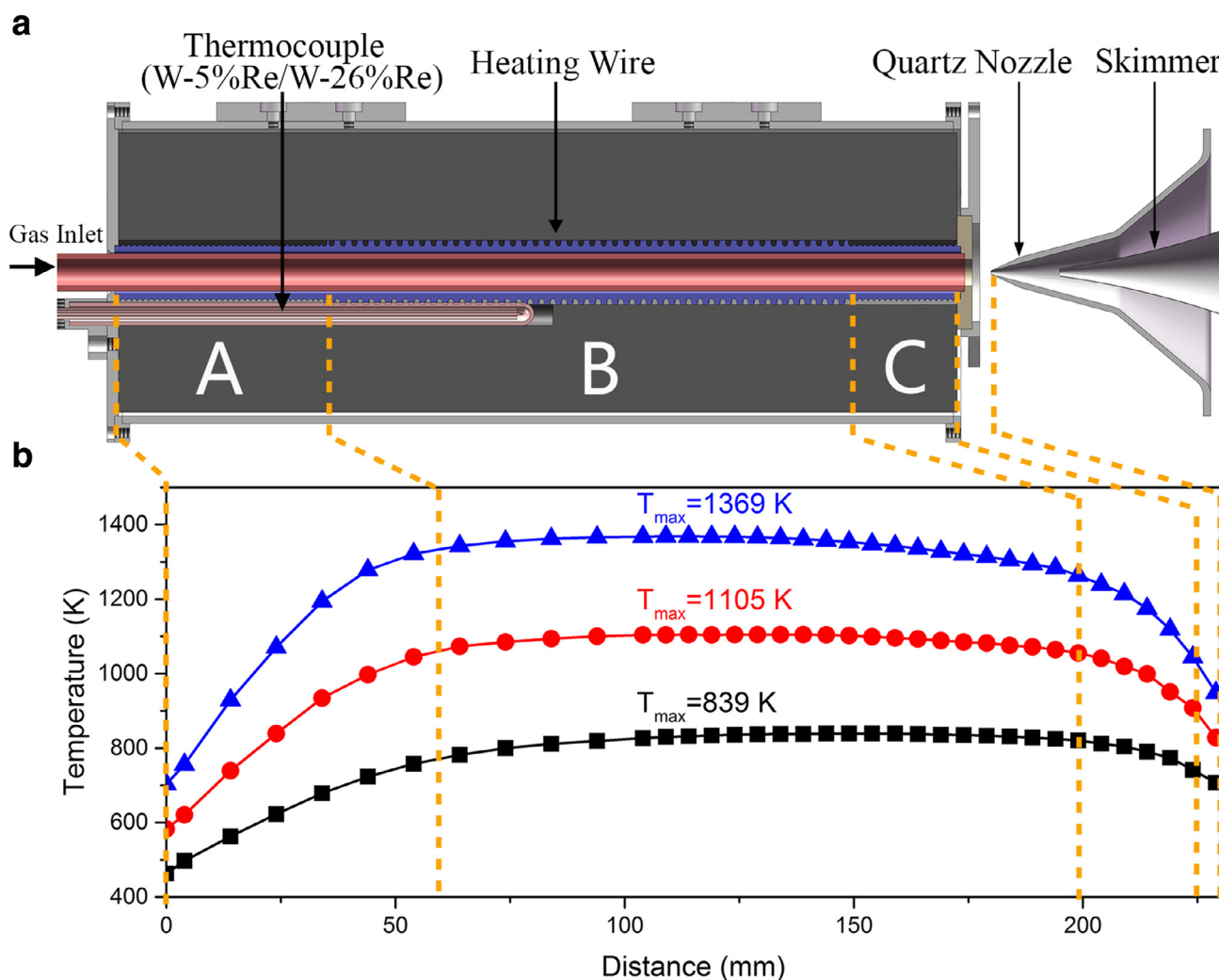


Fig. 1. (a) Schematic diagram of pyrolysis reactor. The W-5%Re/W-26%Re thermocouple shown in the middle region of the heating wire outside the flow tube was used to monitor the furnace temperature during the pyrolysis experiment. (b) The measured temperature profiles along the centerline of flow tube by an S-type thermocouple (not shown).

the pressure in the inner chamber. The combustible gas was spark-ignited and the holes were opened simultaneously. Flame propagation was recorded via a schlieren system with a high-speed camera recording at 9009 fps. The camera resolution was set to 700×350 pixels, while the spatial resolution of the camera corresponded to approximately 0.1 mm/pixel.

The laminar burning velocities of the butane isomers were measured at the initial temperature of 298 K and the initial pressures of 1, 2, 5 and 10 atm using both the SJTU and PU apparatuses. Each measurement was repeated three times. Flame radius ranging from 1.0 to 2.3 cm for the SJTU apparatus and 0.7 to 1.7 cm for the PU apparatus were selected to eliminate the ignition and confinement effects [63,64]. The nonlinear extrapolation method reported by Kelley and Law [22] was used for data processing.

3. Kinetic modeling

In this work, a kinetic model consisting of 148 species and 1174 reactions was developed to interpret the intermediate-to-high temperature combustion chemistry of *n*-butane ($n\text{C}_4\text{H}_{10}$) and *i*-butane ($i\text{C}_4\text{H}_{10}$). The sub-mechanisms of butane isomers were constructed and will be discussed in detail later. The sub-mechanisms of butene and propene (C_3H_6) were adopted from our previous butene and butanol models [51,58,65–67] and the propene model

of Burke et al [68], respectively. The rest of the C_2 – C_4 mechanism, as well as the thermodynamic and transport data, were taken from our *n*-decane model [49], while the C_0 – C_1 mechanism was taken from our recent methanol model [69]. Simulation was performed using the Chemkin-Pro software [70].

3.1. Unimolecular reactions of butane isomers

The unimolecular decomposition of butane isomers involves C–C bond and C–H bond dissociation reactions. Because of the much lower bond dissociation energies, the C–C bond dissociation reactions are considered to be much more important [17,35,43]. There are two types of C–C bond dissociation reactions (R1 and R2) for *n*-butane and only one (R3) for *i*-butane. As shown in Figs. 2 and 3, the rate constants of R1–R3 were investigated by many groups [36,43,71–83]. Among them, Oehlschlaeger et al. [43] first measured the pressure-dependent rate constants of R1–R3 at high temperatures in the shock tube and performed RRKM/master equation calculations to derive the fall-off effect. Their rate constants generally locate in the middle of the literature value distributions and were adopted in the present model. Furthermore, the C–H bond dissociation reactions of the butane isomers were also included in the present model, while the rate constants were taken from the model of Wang et al. [46].

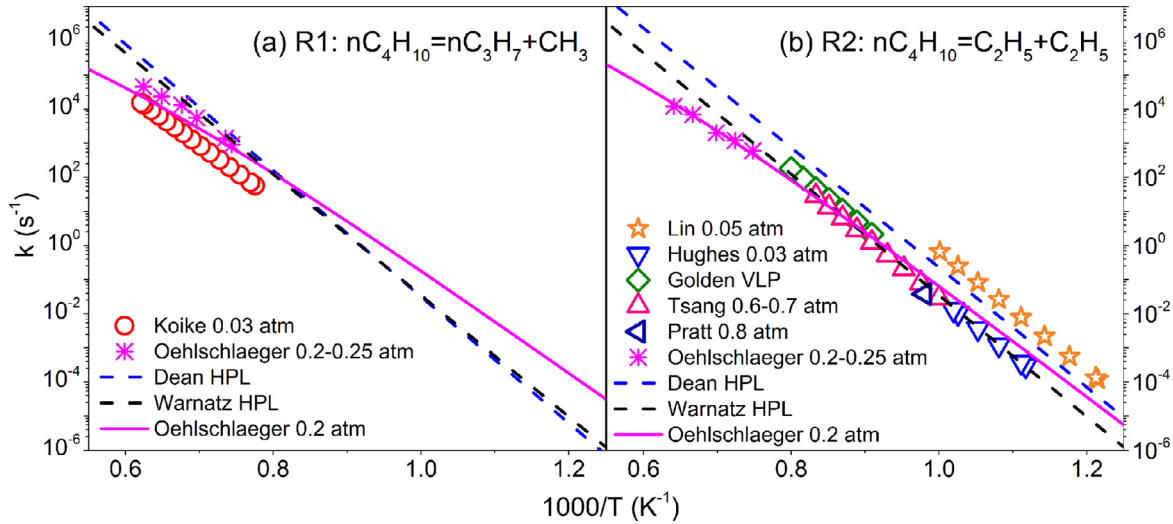


Fig. 2. Comparison of rate constants of (a) R1 and (b) R2 in literature. Symbols, solid lines and dash lines denote the results from experiments [43,71–76], calculations [43] and review papers [77,78], respectively.

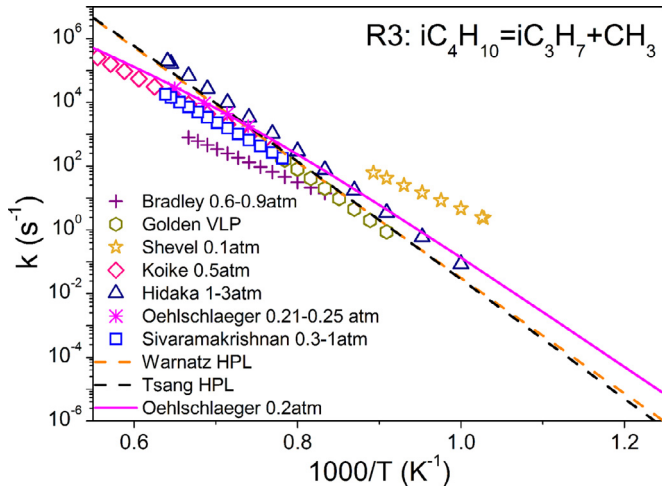
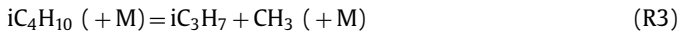
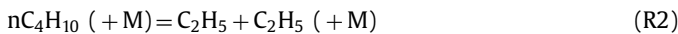
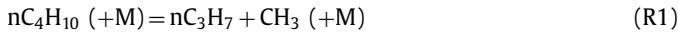


Fig. 3. Comparison of rate constants of R3 in literature. Symbols, solid lines and dash lines denote the results from experiments [36,43,71,79–82], calculations [43] and review papers [78,83], respectively.



3.2. H-atom abstraction reactions of butane isomers

The H-atom abstraction reactions of the butane isomers by radicals (H, CH₃, OH, O and HO₂) were incorporated in the present model. Peukert et al. [84] recently studied the H-atom abstraction reactions of butane isomers by H atom (R4–R7) using shock tube experiments and theoretical calculations, which were adopted in the present model. The H-atom abstraction reactions of *n*-butane by CH₃ radical (R8 and R9) were referred to the site-specific H-atom abstraction reactions by CH₃ radical reported by Orme et al. [85]. Goos et al. [86] investigated the H-atom abstraction reactions of *i*-butane by CH₃ radical (R10 and R11) at 800–950 K and low pressure (80 and 130 mbar) in a quasi-wall-free reactor, which were adopted in the present model. The rate constants of H-atom abstraction reactions by OH radical for *n*-butane (R12 and R13) and

i-butane (R14 and R15) were taken from the experimental work of Droge and Tully [87] and Sivaramakrishnan et al. [88], respectively. Furthermore, the rate constants of H-atom abstraction reactions of the butane isomers by O atom and HO₂ radical were taken from the model of Zhou et al. [89].



3.3. β -Scission reactions of butyl radicals

The four butyl radicals mainly undergo two further classes of reactions, i.e. β -C–C and β -C–H scission reactions. Several previous experimental and theoretical studies have been performed on the decomposition of butyl radicals. For the 1- and 2-butyl radicals (*p*C₄H₉ and *s*C₄H₉), pressure-dependent rate constants of β -C–C and β -C–H scission reactions were taken from the model of Zhou et al. [89]. The β -C–H scission reactions of *i*-butyl (*i*C₄H₉) and *t*-butyl (*t*C₄H₉) radicals were referred to the analogous reactions of *n*-propyl (*n*C₃H₇) and *i*-propyl (*i*C₃H₇) radicals calculated by Miller and Klippenstein [90]. The rate constants of β -C–C scission reaction of *i*-butyl radical were also taken from the model of Zhou et al. [89].

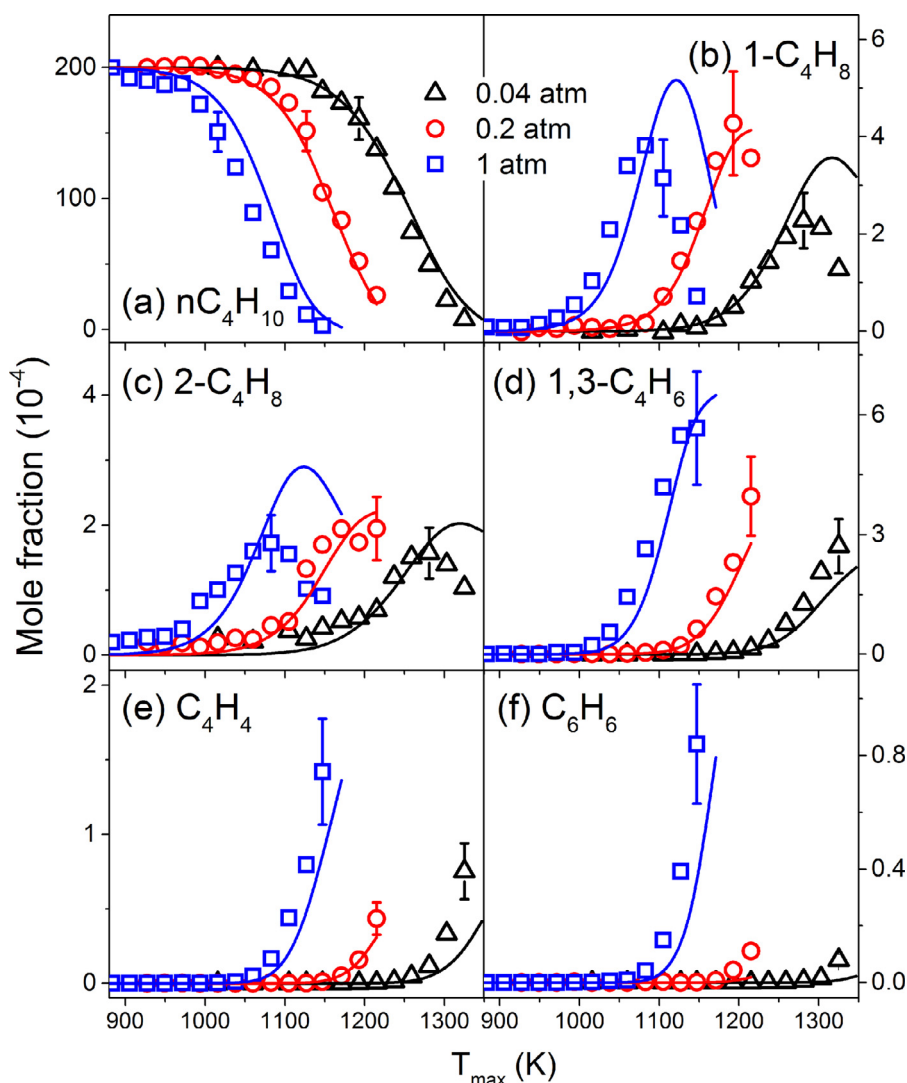


Fig. 4. Experimental (symbols) and simulated (lines) mole fraction profiles of C_4 and C_6 species in n -butane pyrolysis at 0.04, 0.2 and 1 atm.

4. Results and discussion

4.1. Flow reactor pyrolysis

Experimental and simulated mole fractions of the pyrolysis species of n -butane and i -butane are shown in Figs. 4–5 and Figs. 6–7, respectively. As suggested by Rasmussen et al. [91], a flow reactor which satisfies the criterions of the Péclet number (Pe) and the vessel dispersion number (D_{disp}/uL) can be reasonably assumed a plug flow. Under the investigated temperature range at 1 atm in this work, the Péclet numbers of our flow reactor were calculated to be 49–64, which are smaller than that of Harper et al. ($Pe=90$) [92] and close to those of Rasmussen et al. ($Pe=16$ –70) [91] and Shrestha et al. ($Pe=54$) [31]. Based on the work of Rasmussen et al. [91], the vessel dispersion numbers of our flow reactor under the same condition were calculated to be 0.006–0.008, which are smaller than the upper critical limit of 0.01. Therefore the flow reactor pyrolysis experiments were simulated using the Plug Flow Reactor module in the Chemkin-PRO software [70]. In the simulation, the measured centerline temperature profiles were used as the input parameters. It is seen that the present model is able to reproduce the decomposition of butane isomers and the formation of most of the pyrolysis products. The ROP analysis was performed to provide insight into the combustion chemistry of the

butane isomers. The temperatures were selected at 1237 K for the 0.04 atm pyrolysis, corresponding to 58% conversion of n -butane and 64% conversion of i -butane, and 1061 K for the 1 atm pyrolysis, corresponding to 47% conversion of n -butane and 51% conversion of i -butane. At both temperatures, most of the pyrolysis products of n -butane and i -butane were abundantly produced. Based on the ROP analysis, the main reaction networks in n -butane and i -butane pyrolysis at 0.04 and 1 atm are illustrated in Fig. 10.

4.1.1. Primary decomposition of butane isomers

The results of n -butane and i -butane are shown in Figs. 4(a) and 6(a), respectively. According to the ROP analysis, the C–C bond dissociation reactions (R1–R3) consume most of the fuels at initial temperature regions of the fuel decomposition. There are two primary-secondary C–C bonds and one secondary-secondary C–C bond in the n -butane molecule, while the secondary-secondary C–C bond (86.8 kcal/mol [93]) is weaker than the primary-secondary C–C bond (88.9 kcal/mol [93]). The ROP analysis shows that contributions of R1 and R2 to the consumption of n -butane are approximately equal at 0.04 atm. As for i -butane, there is only one primary-tertiary C–C bond dissociation reaction (R3). Figure 8 compares the mole fraction profiles of the butane isomers at different pressures. Both experimental and simulated results indicate that the initial decomposition temperatures of i -butane are slightly

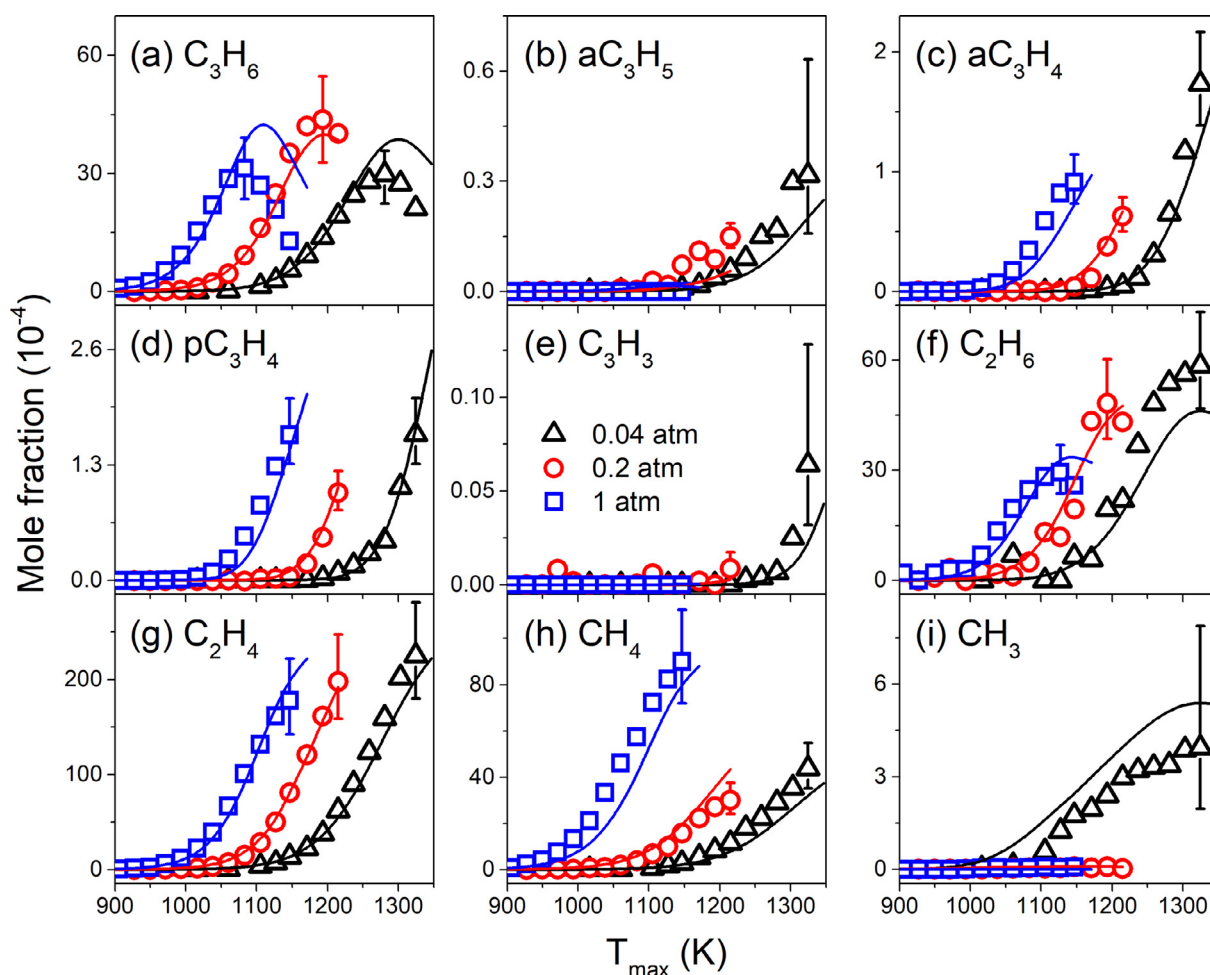


Fig. 5. Experimental (symbols) and simulated (lines) mole fraction profiles of C_1 – C_3 species in n -butane pyrolysis at 0.04, 0.2 and 1 atm.

lower than those of n -butane, as the primary-tertiary C–C bond is more easily to break than the primary-secondary and secondary-secondary C–C bonds in view of the same bond numbers for the isomers.

It is further observed from Fig. 8 that the decomposition temperature regimes of the butane isomers are narrowed as pressure increases, which can be explained by the increasing contributions of the bimolecular reactions. The ROP analysis reveals that H-atom abstraction reactions by H atom and CH_3 radical (R4–R11) play an important role in the decomposition of the butane isomers at relatively high temperatures when enough radicals are produced. Particularly, the H-atom abstraction reactions by CH_3 radical (R10 and R11) to the consumption of i -butane contribute much more than those (R8 and R9) to the consumption of n -butane due to the substantial production of CH_3 radical in the i -butane pyrolysis. For example, R10 and R11 together consume 16% of i -butane at 0.04 atm, while R8 and R9 only contribute 7% to the consumption of n -butane at the same pressure. This also leads to higher production of methane (CH_4) in the pyrolysis of i -butane. Previous studies [93] also demonstrated that the dissociation energies of C–H bond decrease in the order of primary, secondary and tertiary carbon sites, so the H atom on the secondary carbon site can be readily abstracted compared with that on the primary carbon site. As a result, H-atom abstraction reactions on the secondary carbon site (R5 and R9) contribute more to the consumption of n -butane than those on the primary carbon site (R4 and R8). For i -butane, ROP analysis shows that the H-atom abstraction reactions on the primary carbon site (R6, R10) play a similar role in the fuel decom-

position with those on the tertiary carbon site (R7, R11) due to the 9 times of the primary C–H bond to the tertiary C–H bond.

Figure 9 compares the simulated results of the present model and two previous C_0 – C_4 models, i.e. the USC Mech II model [46] and the Aramco 2.0 model [89]. It is observed that the fuel profiles predicted by the previous models shift to higher temperatures at all investigated pressures, ostensibly because these two models were not validated against the pyrolysis data of butane isomers.

4.1.2. Formation and consumption of alkenes and benzene in butane isomers

C_2 – C_4 alkenes were detected in the pyrolysis of both butane isomers, as shown in Figs. 4–7. It is recognized that ethylene (C_2H_4) is the most abundant product in the pyrolysis of n -butane, while propene has comparable concentration levels with ethylene and even early formation temperatures in the pyrolysis of i -butane. This demonstrates the strong influence of the isomeric fuel structure on the formation of key pyrolysis products. According to the ROP analysis, the β -C–C or β -C–H scission reactions of C_2 – C_4 n -alkyl radicals (R16–R18) produced from the fuel decomposition reactions contribute nearly 90% to the formation of ethylene in n -butane pyrolysis. Among the three n -alkyl radicals, n -propyl radical is mainly produced from the primary-secondary C–C dissociation reaction (R1) and 1-butyl radical is dominantly formed from the primary H-atom abstraction reactions (R4 and R8), while ethyl radical (C_2H_5) can be produced from the secondary-secondary C–C bond dissociation reaction (R2) and the β -C–C scission reaction

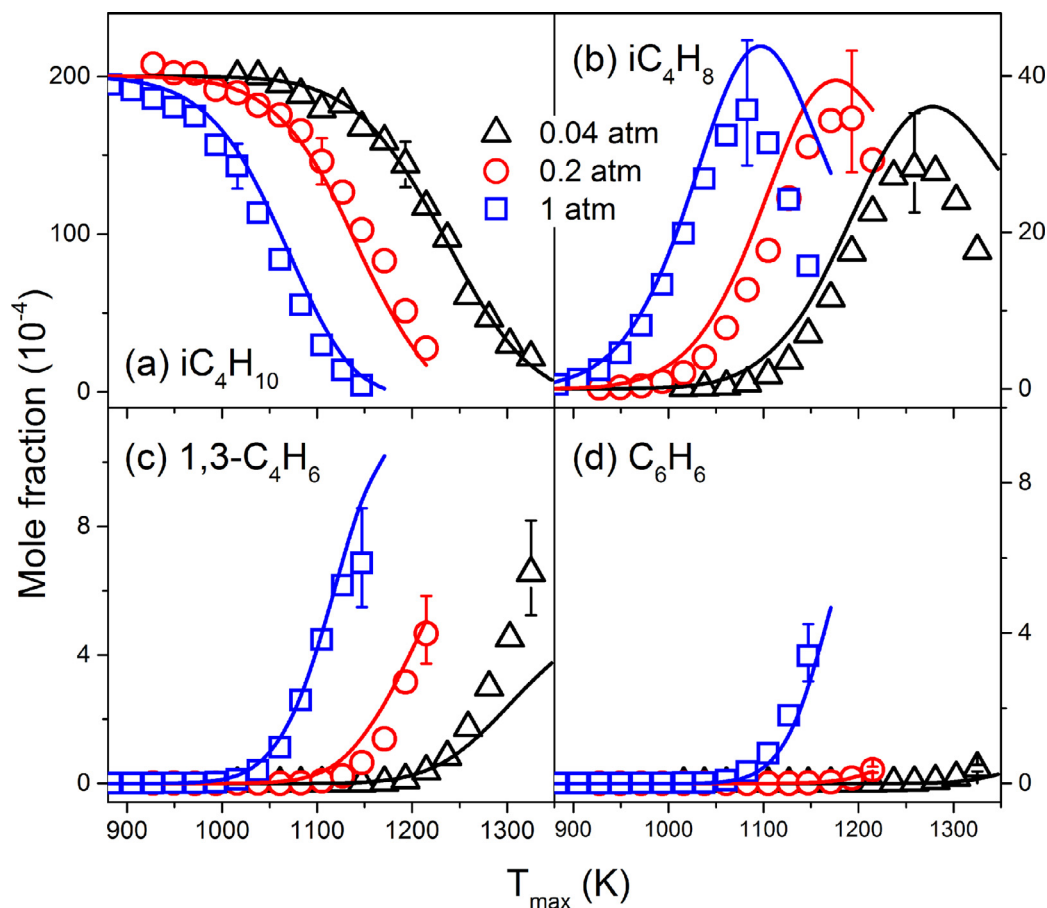


Fig. 6. Experimental (symbols) and simulated (lines) mole fraction profiles of C₄ and C₆ species in *i*-butane pyrolysis at 0.04, 0.2 and 1 atm.

of 1-butyl radical (R16). As a result, ethylene attracts the greatest carbon flux from *n*-butane, as shown in Fig. 10. The formation of propene is less preferred in *n*-butane pyrolysis because its main formation pathway is only the β -C–C scission reaction (R19) of 2-butyl radical that is produced from R5 and R9. Furthermore, the β -C–H scission reactions of 1-butyl and 2-butyl radicals are the dominant formation pathways of 1-butene (1-C₄H₈) and 2-butene (2-C₄H₈) in *n*-butane pyrolysis. However the two reactions are only minor consumption pathways of 1-butyl and 2-butyl radicals, leading to relatively low concentrations of 1-butene and 2-butene, as shown in Fig. 4(b) and (c).



As mentioned above, both ethylene and propene are the dominant alkene products in *i*-butane pyrolysis. As shown in Fig. 10, three major radicals are produced in the initial decomposition of *i*-butane, i.e. *i*-propyl, *i*-butyl and *t*-butyl radicals. The β -C–H scission reaction of *i*-propyl radical (R20) and the β -C–C scission reaction of *i*-butyl radical (R21) are their dominant decomposition pathways and contribute almost 95% together to the formation of propene. Meanwhile the abundance of ethylene is mainly attributed to the decomposition of propene via $C_3H_6 + H = C_2H_4 + CH_3$, which can be verified by the higher formation temperatures of ethylene than propene. Thus, propene be-

comes the dominant fuel-specific product in *i*-butane pyrolysis, which is one of the most significant differences between the pyrolysis of *n*-butane and *i*-butane. Furthermore, *t*-butyl radical is mainly consumed by the β -C–H scission reaction (R22) which is also the dominant formation pathway of *i*-butene (iC₄H₈). The relatively large carbon flux from *t*-butyl radical leads to the abundant formation of *i*-butene, making its maximum mole fractions in *i*-butane pyrolysis several times higher than those of 1-butene and 2-butene in *n*-butane pyrolysis.



The formation of benzene (C₆H₆) is also relevant in the pyrolysis of butane isomers. As seen from Figs. 4(f) and 6(d), *i*-butene pyrolysis produces approximately five times higher concentrations of benzene than *n*-butane pyrolysis at all investigated pressures. As discussed above, propene attracts substantial carbon flux from *i*-butane, leading to much more abundant production of the C₃ precursors of benzene such as allene (aC₃H₄), propyne (pC₃H₄) and propargyl radical (C₃H₃). In particular, propargyl radical is considered as a key precursor of benzene [94] and its more abundant production in *i*-butane pyrolysis explains the higher concentration levels of benzene. Furthermore, no larger aromatic hydrocarbons were observed in the pyrolysis of both butane isomers.

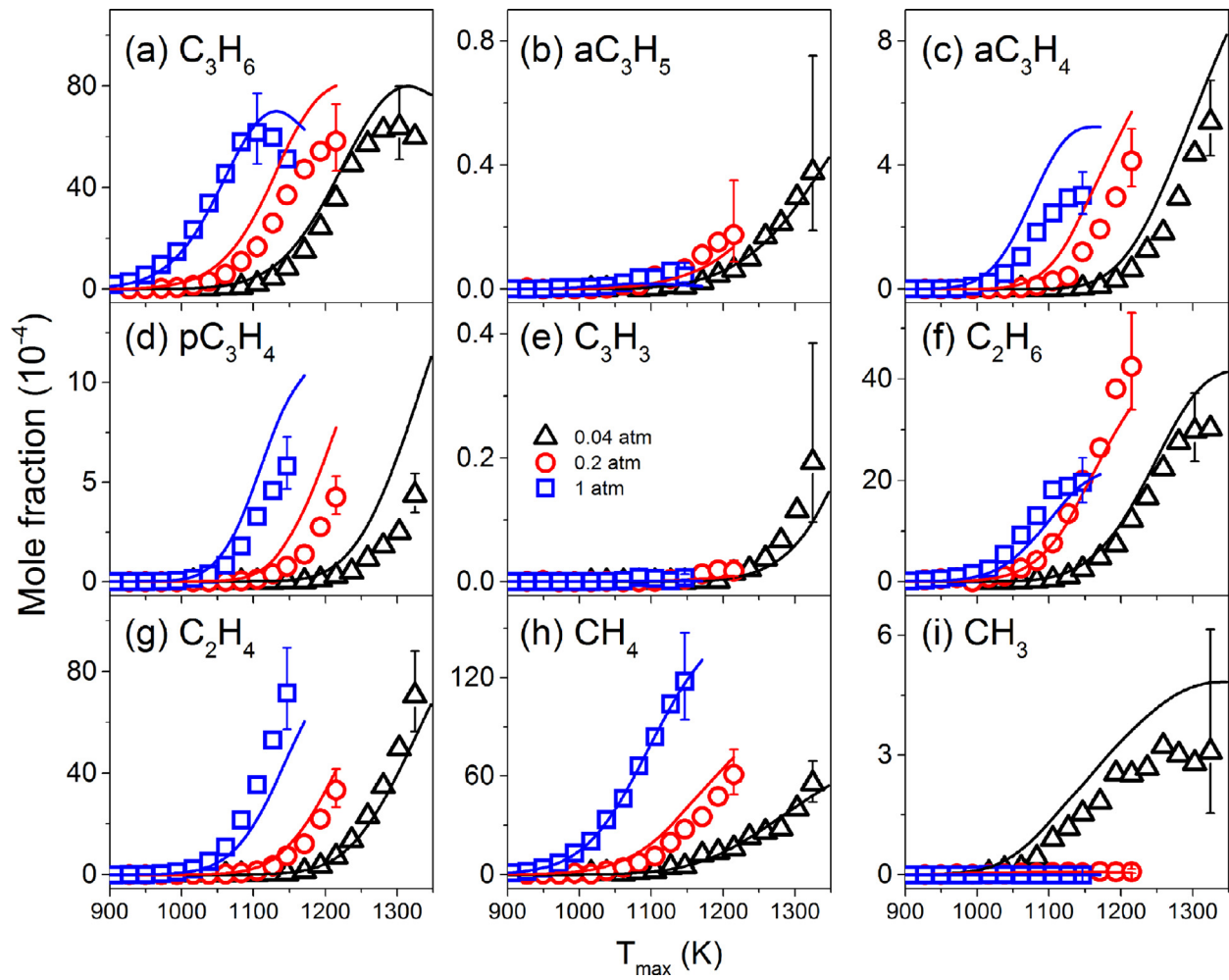


Fig. 7. Experimental (symbols) and simulated (lines) mole fraction profiles of C_1 – C_3 species in *i*-butane pyrolysis at 0.04, 0.2 and 1 atm.

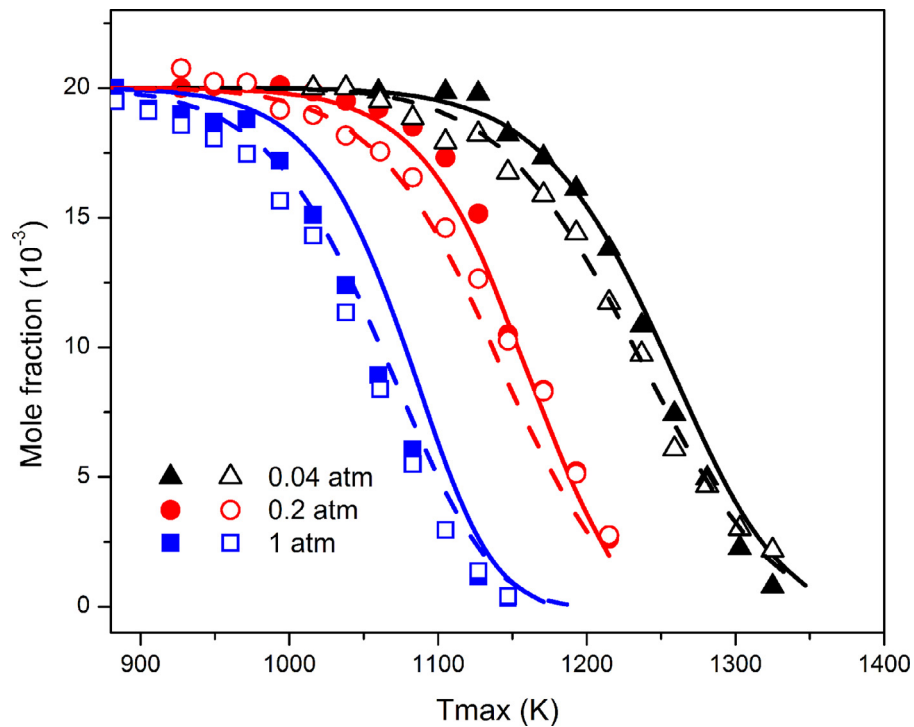


Fig. 8. Comparisons between mole fraction profiles of *n*-butane (experimental results - solid symbols; simulated results - solid lines) and *i*-butane (experimental results - open symbols; simulated results - dash dot lines) at 0.04, 0.2 and 1 atm.

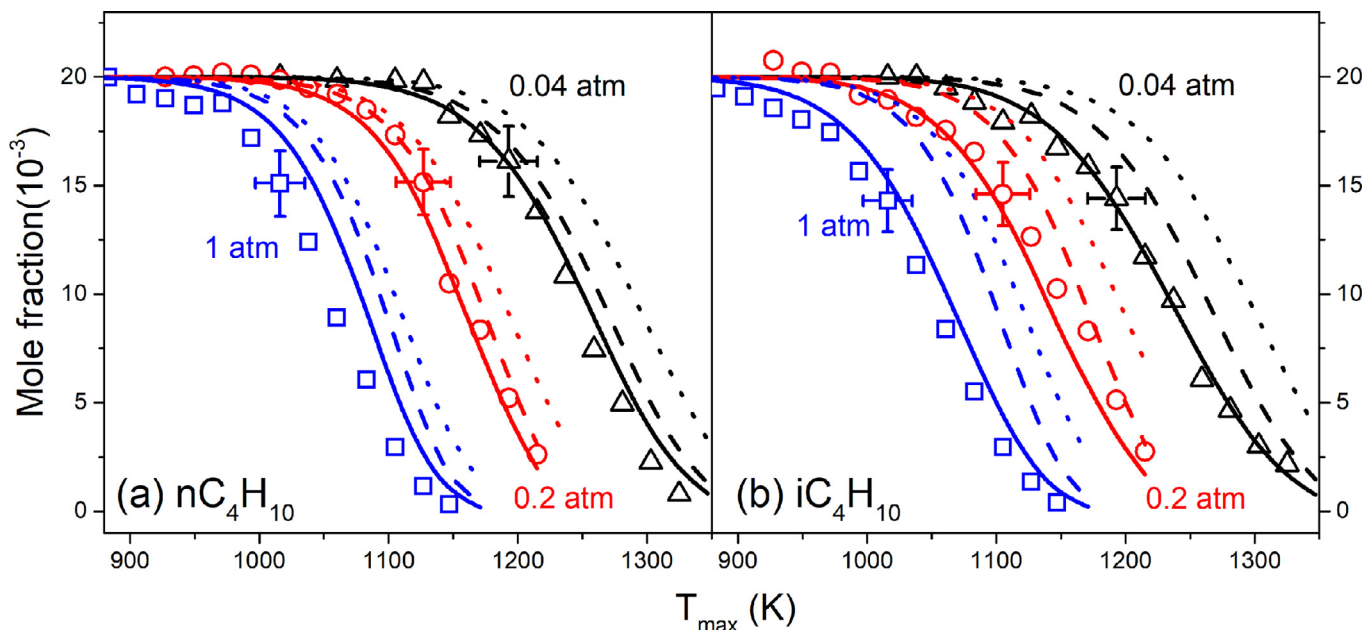


Fig. 9. Comparisons between the experimental (symbols) and simulated mole fraction profiles of (a) n -butane and (b) i -butane. The present model - solid lines; USC Mech II model [46] - dot lines; Aramco 2.0 model [89] - dash lines.

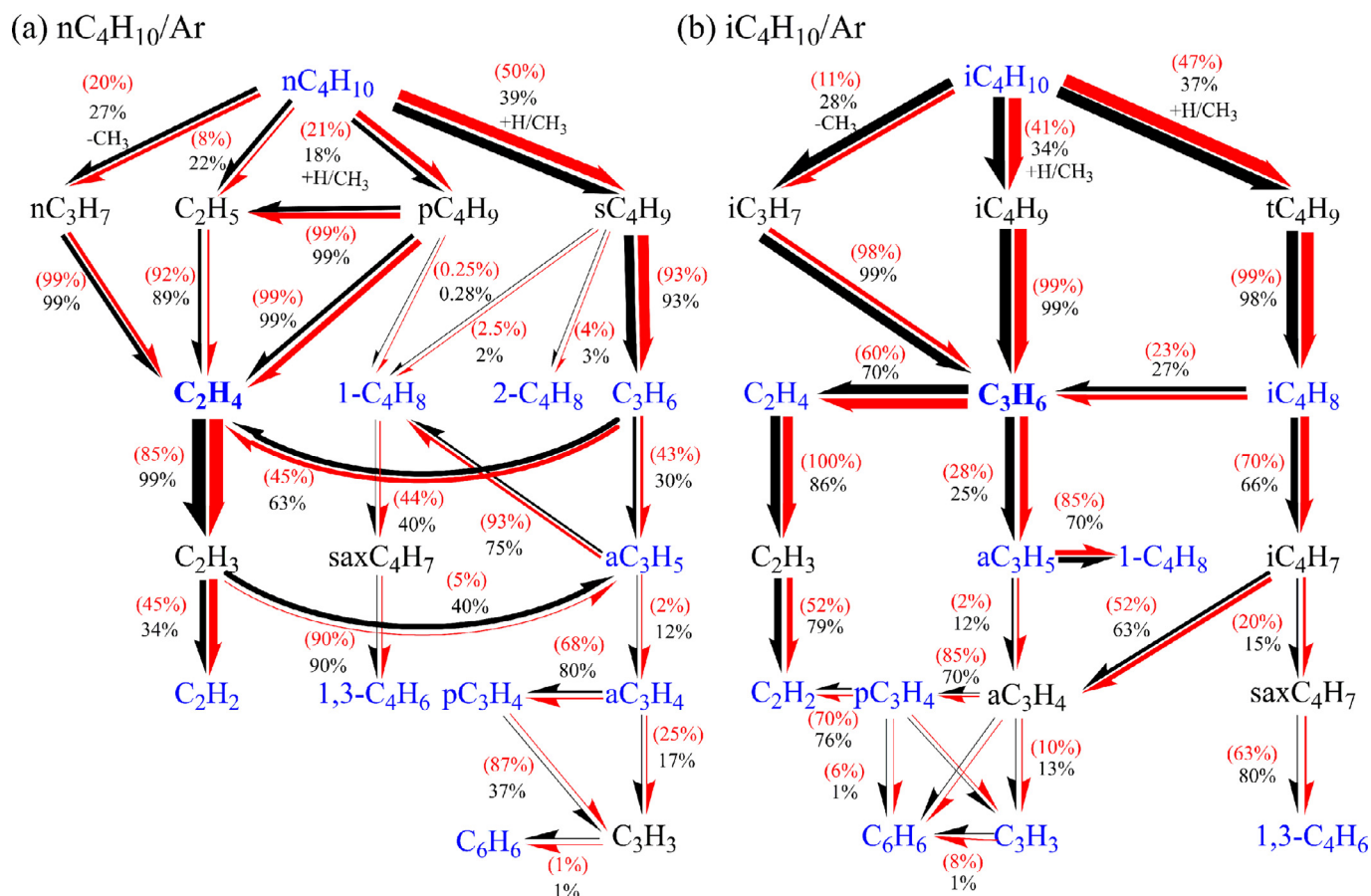


Fig. 10. The reaction networks of (a) n -butane and (b) i -butane pyrolysis at 0.04 atm, 1237 K (black arrows, black numbers) and 1 atm, 1061 K (red arrows, red numbers in parentheses). The numbers denote the percentages of carbon fluxes for corresponding reactions. Species detected in the experiments are marked with blue color. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

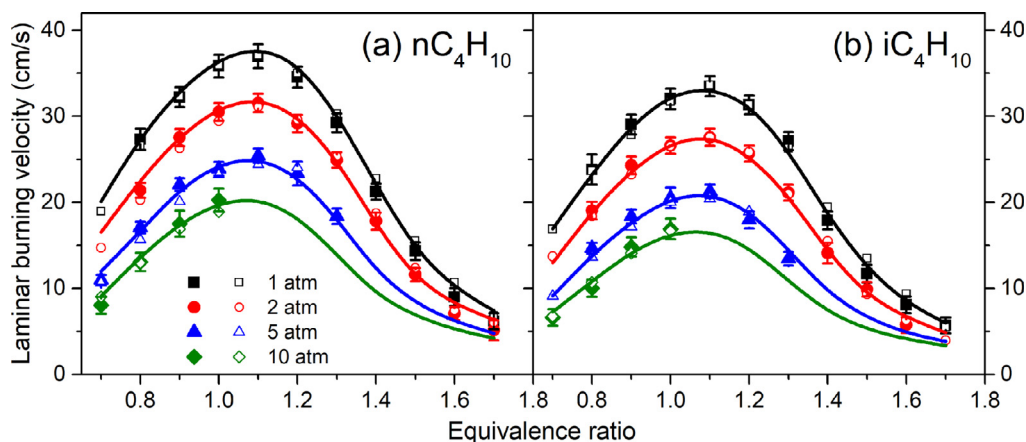


Fig. 11. Comparisons between experimental (solid symbols for the SJTU data and hollow symbols for the PU data) and simulated (lines) laminar burning velocities of (a) *n*-butane/air and (b) *i*-butane/air mixtures at 298 K and 1–10 atm. To clearly show the results, only uncertainties of the SJTU data were plotted.

4.2. Laminar burning velocity

As introduced above, in view of the scarcity of high-pressure experiments, laminar burning velocities of both butane isomers were measured using spherically expanding flames at pressures up to 10 atm in this work. Simulation was performed using the Premixed Laminar Flame Speed Calculation module in the Chemkin-PRO software [70]. Soret effect was considered and mixture-averaged transport was used. Sufficient refinement over a domain of 10 cm (1000 grid points, GRAD 0.1, CURV 0.1) was used to ensure convergence. Figure 11 illustrates the experimental and simulated laminar burning velocities of butane isomers at four pressures (1, 2, 5, and 10 atm), showing that the present model can satisfactorily reproduce the experimental data over the entire equivalence ratios from atmospheric to elevated pressures. Furthermore, as shown in Fig. S1 in the *Supplementary Materials*, good agreement is also observed for butane isomers between the present and previous measurements at 298 K and 1 atm, especially for those measured after 2007.

As shown in Fig. 11, the experimental results from the SJTU apparatus and the PU apparatus agree well. The slight differences are caused by the systematic errors existing in the two different combustion vessel apparatuses. The laminar burning velocities of *n*-butane are always larger than those of *i*-butane at all investigated pressures, and the peaks are located around $\phi = 1.1$ for both isomers. For the 1 and 2 atm experiments, the rich limit is $\phi = 1.7$. Due to the influence of the enhanced flame instabilities at elevated pressures, the maximum equivalence ratios are 1.3 and 1.0 for 5 and 10 atm experiments, respectively. Furthermore, laminar burning velocities of both butane isomers decrease with the increasing initial pressure. This trend is mainly due to the corresponding increase in the density. In fact, the mass burning flux, which is the density-weighted burning velocity, increases with increasing pressure, as is reasonable to expect because of the overall increase in reactivity with the increasing pressure.

Figure 12 shows the most sensitive reactions to the laminar burning velocities of *n*-butane and *i*-butane at 1 and 10 atm under lean ($\phi = 0.7$) and rich ($\phi = 1.7$) conditions, noting that reactions involved in the C_0 – C_1 , C_2 , C_3 and C_4 sub-mechanisms are indicated in black, blue, olive and red, respectively, and that $CH_3 + CH_3 (+M) = C_2H_6 (+M)$ is treated as a reaction in the C_0 – C_1 sub-mechanism. It is clearly seen that the C_0 – C_1 core mechanism undoubtedly plays a vital role in enhancing or inhibiting flame propagation, and every single reaction indicated in black shows similar performance on flame propagation for *n*-butane and *i*-butane. The most important chain branching reaction R23 exhibits the

strongest promoting effect, followed by the chain propagation reaction R24 which is also the most important reaction for heat release in hydrocarbon flames. As the equivalence ratio increases from 0.7 to 1.7 at both pressures for *n*-butane and *i*-butane, the sensitivity coefficients of R23 increase dramatically while those of R24 have the opposite trend, as the consequence of the fact that rich flames are abundant in H atom but deficient in OH radical and O atom, while lean flames are abundant in OH radical and O atom but less abundant in H atom.



Figure 12 shows that the two combination reactions involving H atom (R25 and R26) have large negative sensitivity coefficients under lean and rich conditions, respectively, which is mainly because reactions competing with R23 for H atom inhibit the combustion process [95]. These two reactions also exhibit strong dependence on the equivalence ratio for both isomers. That is, as the equivalence ratio changes from 0.7 to 1.7 at 1 and 10 atm, the sensitivity coefficients of R26 increase substantially while those of R25 decrease dramatically. It is also noted that R25 has a positive sensitivity coefficient at 10 atm and $\phi = 1.7$, indicating a strong pressure effect. Such phenomenon was also observed in our recent methanol/air flame investigation [69]. The reason is that HO_2 radical is abundantly produced under such high-pressure and very rich conditions, triggering the chain branching sequence ($HO_2 \rightarrow H_2O_2 \rightarrow 2OH$) that can promote flame propagation.



In addition to the small species reactions, fuel-specific reactions (R2–R7) are also among the most sensitive reactions for butane isomers, especially under rich and high-pressure conditions. As shown in Fig. 12, the unimolecular decomposition reactions of *n*-butane (R2) and *i*-butane (R3) both have positive sensitivity coefficients under these conditions since their products, ethyl and *i*-propyl radicals, can readily convert to H. In particular, the β -C–H scission of the ethyl radical (R18) is also among the most sensitive reactions for *n*-butane, indicating how the primary and secondary decomposition reactions of *n*-butane influence its flame propagation. For H-atom abstraction reactions (R4–R7), their roles are different in flame propagation, i.e. R4 producing 1-butyl

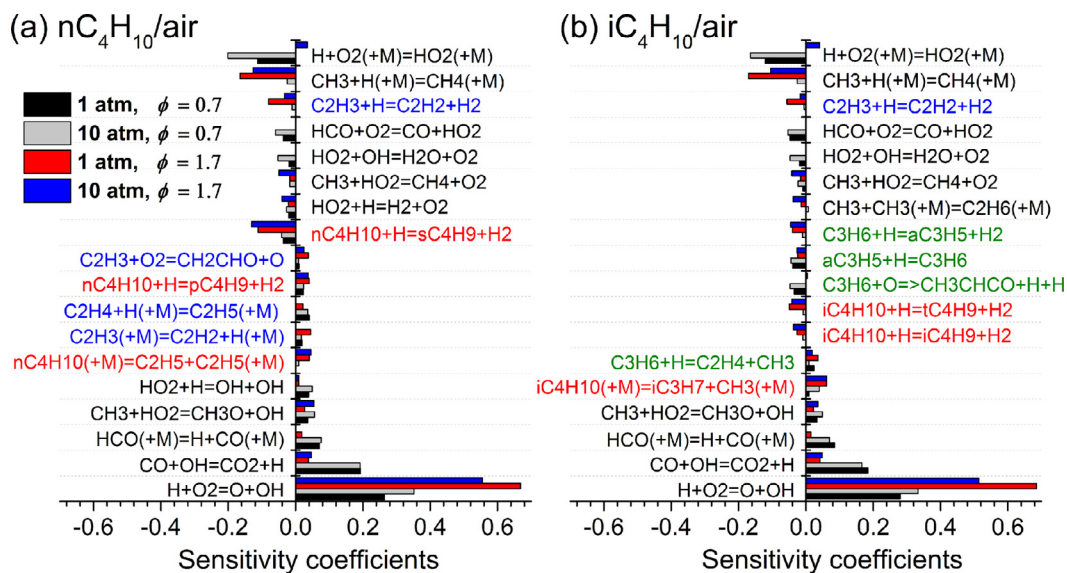


Fig. 12. Sensitivity analysis of (a) *n*-butane and (b) *i*-butane laminar burning velocities at 298 K. The reactions marked as black, blue, olive and red indicate the C₀–C₁, C₂, C₃ and C₄ reactions respectively. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

radical has positive sensitivity coefficients while R5–R7 producing 2-butyl, *i*-butyl and *t*-butyl radicals have negative values. The reason is that alkenes produced by butyl radicals through β -scission reactions have apparent influence on flame propagation [95]. That is, propene produced from 2-butyl and *i*-butyl decomposition and *i*-butene produced from *t*-butyl decomposition tend to produce resonantly stabilized allyl (aC_3H_5) and *i*-butenyl (iC_4H_7) radicals, and consequently inhibit flame propagation for the butane isomers, while ethylene produced from 1-butyl decomposition can only form reactive vinyl radical (C_2H_3) and eventually enhance the flame propagation of *n*-butane. As discussed above, ethylene and propene are the most preferred products in *n*-butane and *i*-butane pyrolysis, respectively. Previous research by Oßwald et al. [41] also drew the similar conclusion in rich premixed flames of butane isomers. As a result, more reactions in the C₂ sub-mechanism are among the most sensitive reactions in *n*-butane flames, while more reactions in the C₃ sub-mechanism are among the most sensitive reactions in *i*-butane flames, as shown in Fig. 12. In summary, the easy production of H atom from the fuel decomposition and the dominance of reactive C₂ chemistry can explain the larger laminar burning velocities of *n*-butane than those of *i*-butane at all investigated pressures.

To further elucidate the effects of elevated pressure on the laminar burning velocities of *n*-butane and *i*-butane, the pressure dependent coefficients β were calculated according to the power-law equation $S_L = S_{L0}(\frac{P}{P_0})^\beta$ [96,97], where P_0 is the reference pressure (1 atm in this work) and S_{L0} is the laminar burning velocity at P_0 . Figure 13 shows the results of β at different equivalence ratios. Due to the limited data under very rich conditions at 5 and 10 atm, β values were calculated with equivalence ratios from 0.7 to 1.3. Results of both the SJTU and PU apparatuses reveal that the absolute values of β for *i*-butane/air flames are always greater than those for *n*-butane/air flames under all investigated equivalence ratios, indicating that *i*-butane/air flames have stronger pressure dependence than *n*-butane/air flames. According to Egolfopoulos and Law [97], termination reactions generally tend to reduce the global reaction order n because of their negative influence on the progress of the global reaction. Consequently *i*-butane/air flames have lower n values than *n*-butane/air flames due to the enhanced role of termination reactions in *i*-butane/air flames shown in Fig. 12. This explains the larger absolute values of

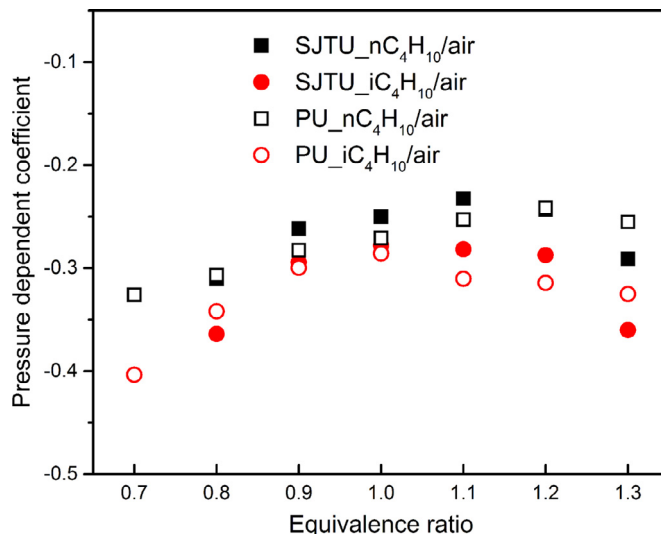


Fig. 13. Pressure dependent coefficients β of *n*-butane/air and *i*-butane/air flames with various equivalence ratios calculated from measured laminar burning velocities.

β and stronger pressure dependence for *i*-butane/air flames considering the relationship $\beta = \frac{n}{2} - 1$ [96,97].

4.3. Validations of literature data

As summarized in Table 1, fundamental combustion experiments on *n*-butane and *i*-butane have been investigated extensively, yielding results that could be used to validate the present model. Particularly, in order to probe the influence of the fuel isomeric structures in combustion chemistry, special attentions were paid to the literature results with identical experimental conditions for the butane isomers. We therefore present in the following sections the validations against the speciation data in the JSR oxidation reported by Dagaut et al. [37,39] and the laminar premixed flames reported by Oßwald et al. [41], as well as the ignition delay time data reported by Ogura et al. [5], with the additional validations given in the Supplementary Materials.

Table 1A list of experiment data of *n*-butane and *i*-butane used to validate the present model.

Experiment type	Temperature	Pressure	Equivalence ratio	Reference
<i>n</i> -butane				
Flow reactor pyrolysis and oxidation	840–1350 K	0.04,0.2,1 atm	∞	This work
	1172 K	1 atm	1.25	[33]
	1100 K	1 atm	∞	[31]
Shock tube pyrolysis and oxidation	1200–1400 K	1.45–1.64 atm	∞	[34]
	1500–1800 K	2.02–2.16 atm	1	[35]
	1050–1250 K	1 atm	1	[37]
JSR oxidation	900–1200 K	1.2,5,10 atm	0.15–4.0	[38]
Laminar premixed flame	260–2100 K	40 mbar	1.71	[41]
	500–1600 K	1 atm	2.6	[42]
	300–2500 K	30 Torr	1.7	[40]
Ignition delay time	1240–1400 K	10 atm	1	[16]
	1300–1700 K	1–4 atm	1	[15]
	1200–1600 K	1.9–2.4 atm	0.72	[5]
	690–1430 K	1–45 atm	0.3–1	[9]
	1000–1600 K	1–21 atm	0.3–2.0	[8]
	1100–1600 K	0.2–2 MPa	0.5,1,2	[11,12]
	1200–1600 K	1.2, 5.3 atm	1	[13]
	$T_u = 298$ K	1.2,5,10 atm	0.7–1.7	This work
Laminar burning velocity	$T_u = 298$ K	1 atm	0.6–2	[18–29]
<i>i</i> -butane				
Flow reactor pyrolysis	840–1350 K	0.04,0.2,1 atm	∞	This work
JSR oxidation	1050–1250 K	1 atm	1.5	[39]
Laminar premixed flame	300–2200 K	40 mbar	1.71	[41]
Ignition delay time	1300–1800 K	1.5–10 atm	0.25,0.5,1,2	[17]
	1200–1600 K	1.9–2.4 atm	0.72	[5]
	1000–1600 K	1–21 atm	0.3–2.0	[8]
	590–1567 K	1–30 atm	0.3,0.5,1,2	[10]
	$T_u = 298$ K	1.2,5,10 atm	0.7–1.7	This work
Laminar burning velocity	$T_u = 298$ K	1 atm	0.6–2	[18,24,26]

4.3.1. JSR oxidation

The present model was validated against the JSR oxidation data reported by Dagaut et al. [37,39]. The experimental conditions were identical for *n*-butane and *i*-butane, that is, 0.22%butane/ O_2/N_2 at 1 atm, 1000–1450 K, residence time of 0.16 s and equivalence ratios of 0.5, 1.0 and 1.5. Mole fraction profiles of stable species were measured as a function of temperature using GC. Simulation was performed using the Perfectly Stirred Reactor module in the Chemkin-PRO software [70]. Limited to the experimental results provided in the original work [37,39], comparison between the experimental and simulated mole fraction profiles for *n*-butane and *i*-butane was performed at the equivalence ratios of 1.0 and 1.5, respectively, as shown in Fig. 14. The simulated results of the USC Mech II model [46] and the Aramco 2.0 model [89] are also compared with those of the present model, as shown in Fig. S11 in the *Supplementary Materials*. Based on the ROP analysis, H-atom abstraction reactions by H, CH_3 , O, OH and HO_2 radicals are dominant pathways for the consumption of both *n*-butane and *i*-butane. Among them, those by H atom are most important while those by OH also show significant contributions. Furthermore, C–C bond dissociation reactions of butane isomers cannot be ignored even under such oxidation conditions. The ROP analysis shows that R1 and R2 contribute 20% to the fuel consumption in the stoichiometric oxidation of *n*-butane, while R3 contributes 32% to the fuel consumption in the rich oxidation of *i*-butane.

4.3.2. Laminar premixed flame

Oßwald et al. [41] investigated the rich ($\phi = 1.71$) laminar premixed flames of both butane isomers (15.7% C_4H_{10} /59.5% O_2 /24.8%Ar) at 40 mbar. Mole fraction profiles of the flame species up to C_7 were measured as function of the burner position using EIMS and SVUV-PIMS. Simulation was conducted using the Premixed Burner Stabilized Flame module in the Chemkin-PRO software [70] with consideration of Soret effect and mixture-averaged transport. The flame temperature profiles

measured by Oßwald et al. [41] were used as input parameters, which was also adopted by Oßwald et al. in their simulation work. Figure 15 presents the experimental and simulated mole fraction profiles of seven major species (butane, O_2 , Ar, H_2O , H_2 , CO and CO_2) in both *n*-butane and *i*-butane flames. The experimental and simulated results of other flame species, as well as the comparison of simulated results by the present model and the model from Oßwald et al. [41], are provided in the *Supplementary Materials*. It is observed that the present model can reproduce the reactant consumption and the product formation. The ROP analysis shows that C–C bond dissociation reactions of *n*-butane (R1 and R2) and *i*-butane (R3) both contribute approximately 10% to the fuel consumption and the rest is contributed by H-atom abstraction reactions. Among the H-atom abstraction reactions, those by H atom are more favored than those of OH in both *n*-butane and *i*-butane flames due to the very rich reaction circumstance. Similar to the situation in pyrolysis, C_2 chemistry and C_3 chemistry dominate further decomposition processes in the rich premixed flames of *n*-butane and *i*-butane, respectively.

4.3.3. Ignition delay time

Ignition delay times of both butane isomers were measured by many groups using shock tubes and rapid compression machines under a wide range of temperature, pressure and equivalence ratio conditions. In particular, Ogura et al. [5] measured the ignition delay times of *n*-butane/*i*-butane mixtures with five different ratios (1.0/0.0, pure *n*-butane; 0.7/0.3; 0.5/0.5; 0.3/0.7; 0.0/1.0, pure *i*-butane) at 1.9–2.4 atm, 1200–1600 K and $\phi = 0.72$ behind the reflected shock waves. The experimental data of pure *n*-butane and *i*-butane and the simulated results using the present model are shown in Fig. 16. The simulated results using the USC Mech II model [46] and the Aramco 2.0 model [89] are also compared with those of the present model, as shown in Fig. S19 in the *Supplementary Materials*. Simulation was conducted using the Closed Homogeneous Batch Reactor module in the Chemkin-PRO software [70] based on the adiabatic and constant-volume assumptions. The

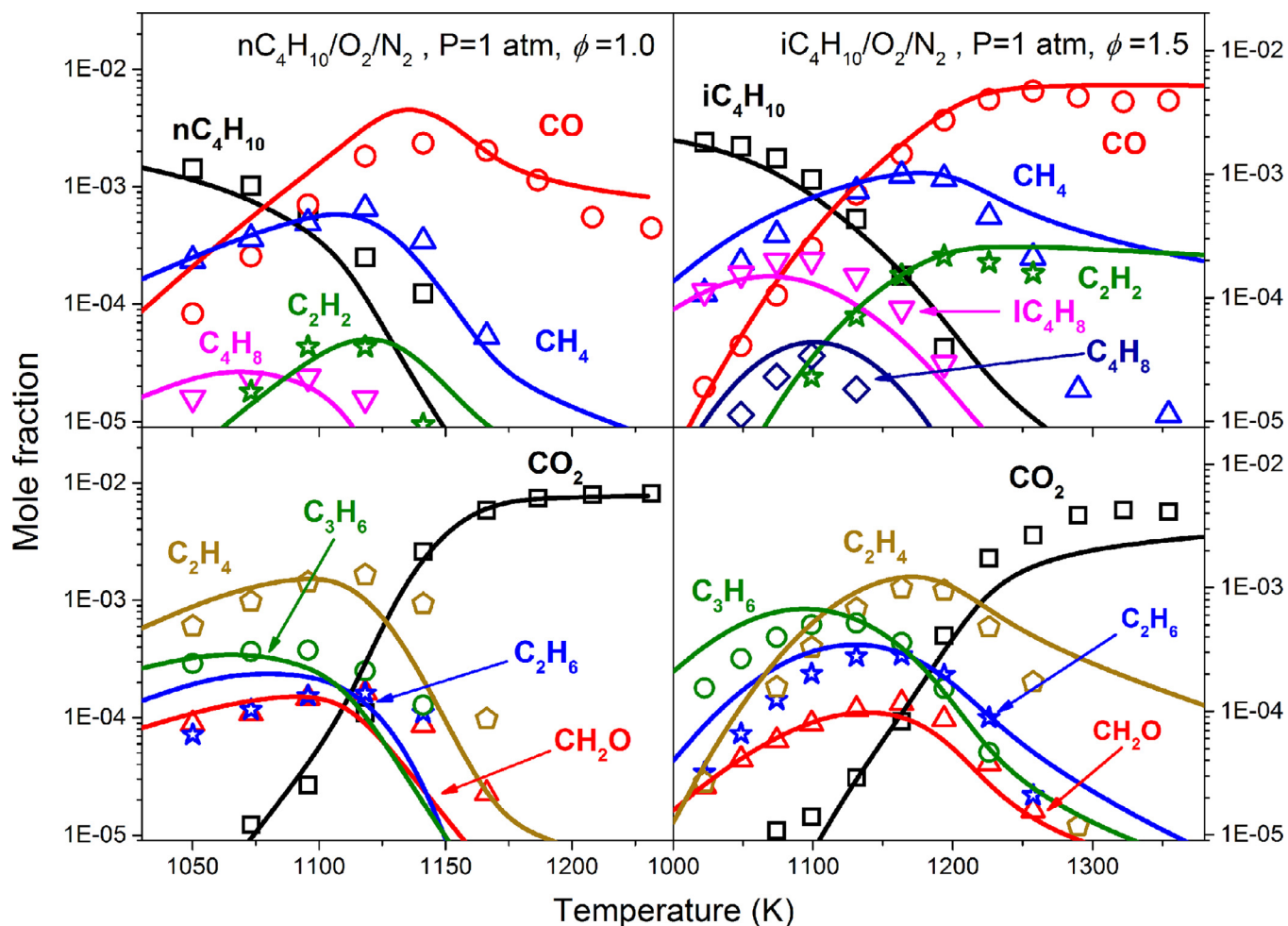


Fig. 14. Experimental (symbols) [37,39] and simulated (lines) mole fraction profiles in the JSR oxidation of n -butane (left) and i -butane (right).

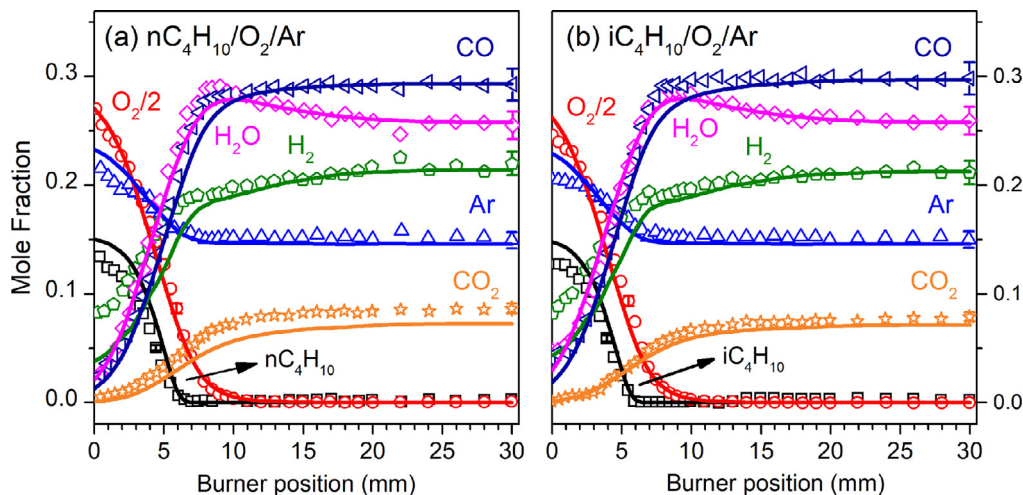


Fig. 15. Experimental (symbols) [41] and simulated (lines) mole fractions of major species in the rich ($\phi = 1.71$) laminar premixed flames of (a) n -butane and (b) i -butane.

ignition delay time τ_{ign} is defined as the time when the pressure rises sharply with an induction time, which was verified to reach a good agreement with the definition of UV emission by Ogura et al. [5]. The present model shows generally good performance in capturing the experimental results. It can be observed that i -butane is slower to be ignited than n -butane, which indicates that the oxidation of i -butane proceeds slower than that of n -butane under the investigated conditions. Similar behavior was also observed in

Fig. 11 that the laminar burning velocities of i -butane are lower than those of n -butane. Furthermore, comparison of the experimental and simulated results for the n -butane/ i -butane mixtures with ratios of 0.7/0.3, 0.5/0.5 and 0.3/0.7 is provided in Fig. S20 of the Supplementary Materials. As expected, a gradual increase of ignition delay times of n -butane/ i -butane mixtures is observed with the increasing i -butane ratio, which is well captured by the present model.

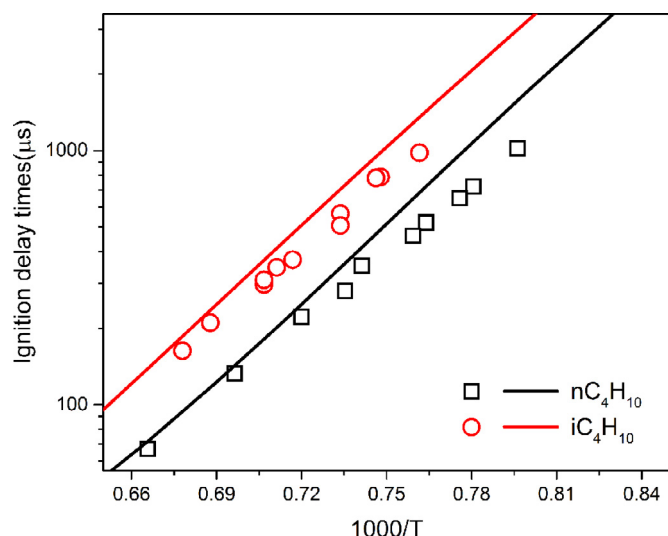


Fig. 16. Experimental (symbols) [5] and simulated (lines) ignition delay times of *n*-butane/O₂/Ar and *i*-butane/O₂/Ar.

5. Conclusions

The pyrolysis and laminar burning velocities of *n*-butane and *i*-butane were investigated in this work. The flow reactor pyrolysis of two butane isomers was performed at 0.04, 0.2 and 1 atm using SVUV-PIMS. Laminar burning velocities of *n*-butane and *i*-butane were measured at 298 K and pressures of 1, 2, 5 and 10 atm using both single- and dual-chamber combustion vessels, which show satisfactory agreement with each other. Different behaviors in both pyrolysis and laminar flame propagation of *n*-butane and *i*-butane were also revealed. A kinetic model for butane isomers was proposed and validated against the present data and the literature data of butane combustion, including the flow reactor pyrolysis and oxidation, the shock tube pyrolysis and oxidation, the JSR oxidation, the laminar premixed flames and the ignition delay times.

Modeling analyses including ROP analysis and sensitivity analysis were performed to facilitate understanding the effects of fuel structure and pressure on the combustion characteristics between *n*-butane and *i*-butane. In the flow reactor pyrolysis, the decomposition of fuel and the distribution of the pyrolysis products are both influenced by the isomeric fuel structure. The weaker primary-tertiary C–C bond than primary-secondary and secondary-secondary C–C bonds leads to faster unimolecular decomposition of *i*-butane than *n*-butane, making the initial decomposition of *i*-butane occur at lower temperatures than those of *n*-butane. The branched structure of *i*-butane also makes its unimolecular decomposition and H-atom abstraction both eventually favor the formation of propene, while most of carbon flux flows from *n*-butane to ethylene. This explains the higher concentrations of benzene in *i*-butane pyrolysis than those in *n*-butane pyrolysis. For the laminar burning velocity, besides key reactions in C₀–C₁ mechanism, a series of reactions involving C₂ and C₃ species are also among the most sensitive reactions in *n*-butane and *i*-butane flames, respectively. The larger laminar burning velocities of *n*-butane than those of *i*-butane are mainly attributed to the easy production of H atom from *n*-butane decomposition and the dominance of the reactive C₂ chemistry. Furthermore, the *i*-butane/air flames bear stronger pressure dependence than the *n*-butane/air flames, which mainly results from the enhanced role of the termination reactions in the *i*-butane flames.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.combustflame.2018.01.002](https://doi.org/10.1016/j.combustflame.2018.01.002).

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